

TX audio source — the radio's codec mic, a PC mic, or digital-mode audio.

Lyra lets you choose what drives the transmit chain, at **Settings** → **TX** → **Mic source**:

- **Mic In** — a microphone in the **Hermes Lite 2's mic jack** (its onboard codec, AK4951 on HL2+). *The default for voice.*
- **PC Soundcard (VAC1)** — audio from your **PC** (a USB/headset mic, or a program) via a virtual audio cable.
- **TCI (digital modes)** — audio streamed from a digital-mode program (MSHV, JTDX, WSJT-X, FLDigi...) over Lyra's TCI link; the mic is bypassed.

(Line In and VAC2 appear greyed out — planned for a later version.) Only one source is live at a time — whatever's selected goes on the air.

Which one:

- Voice with a mic on the radio → **Mic In**.
- Voice with a PC/USB mic or headset → **PC Soundcard (VAC1)**.
- FT8 / FT4 / RTTY / digital → **TCI (digital modes)**.

Watch the ALC meter while setting any source. A little ALC action on peaks is good; ALC slamming hard the whole time means too much input — back the gain off.

Mic In (radio's codec mic, AK4951)

1. Plug your mic into the **MIC jack on the Hermes Lite 2**.
2. **Settings** → **TX** → **Mic source** → **Mic In**.
3. Key into a dummy load, speak at your normal level, and raise **Mic Gain **until the ALC** meter just flicks on voice peaks.**
4. If your mic is genuinely weak, tick **Mic Boost (+20 dB hardware, codec mic only)** — the HL2 codec's analog preamp — then fine-trim with Mic Gain. *(Mic Boost only affects the codec mic; it does nothing for PC or TCI sources.)*

PC Soundcard (VAC1) — a PC/USB mic

Use this when your microphone is on the computer. It rides the same virtual-cable bridge as digital modes (install one, e.g. VB-CABLE, if you're routing from another program; for a plain PC/USB mic, point VAC1's input straight at the mic).

1. **Settings** → **TX** → **Mic source** → **PC Soundcard (VAC1)**.
2. **Settings** → **Audio** → **Virtual Audio Cable (VAC1)**: tick **Enable**, set **Input device** to your PC mic (or the cable's output side if feeding from another program), and set **TX gain** so the ALC sits right when you talk. **Combine input** mixes a two-channel source to mono if it comes through one-sided.
3. Key up and check the ALC as above.

TCI (digital modes)

For FT8/FT4/RTTY/etc., the digital program sends transmit audio to Lyra over TCI — no microphone involved.

1. **Settings** → **TX** → **Mic source** → **TCI (digital modes)**.
2. **Settings** → **Network** — make sure the TCI server is enabled.
3. In the program (MSHV / JTDX / WSJT-X / FLDigi...), select **TCI** and point it at Lyra's address + TCI port.
4. Set levels in the **program** so Lyra's ALC shows only light action — digital wants clean, just-below-ALC drive. If it's hot, Lyra also has a TCI input-gain trim to tame an over-driven client.

Setting levels (any source)

Aim for voice peaks (or digital drive) to push the **ALC** into a little action, not pinned. Too hot (ALC slammed) = distortion/splatter — reduce the gain (Mic Gain, VAC1 TX gain, or the digital app's level). Too cold (no ALC, low power) = raise it. For voice, start on a **dummy load** and use

MON TX to hear your own processed audio while you dial it in.

If something's wrong

- **No transmit audio:** confirm **Mic source** matches what you're using

(a PC mic won't work if the source is still *Mic In*), you're keyed, and PA + Drive are set for real RF.

- **Mic In weak even at high Mic Gain:** tick **Mic Boost (+20 dB)**, then

trim with Mic Gain.

- **Distorted, ALC pinned:** back off the gain for the active source.

- **PC mic silent:** source must be **PC Soundcard (VAC1)** *and* VAC1

enabled with the right **Input device**; enable **Combine input** if a stereo source is one-sided.

- **Digital app keys but nothing goes out:** source must be **TCI**, the

app set to TCI on the right port, and the app's output level up.
